Linear functions — Part 2

Algebra · Math · 85 questions

2.1 17c37dcf-800e-41d5-8367-1ccfbfa33af2 · Medium

The pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch psi. For each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. What is the total pressure, in psi, exerted on the scuba diver at 105 feet below sea level?

A. 60.90

B. 31.50

C. 14.70

D. 0.44

2.2 f8b5e794-72b0-4393-8d1c-cd433c069575 · Medium

The function f is defined by $fx = \frac{9}{7}x + \frac{8}{7}$. For what value of x does $fx = 5$?

Student-produced response: _____

2.3 d9f57fdb-9af6-41d0-9c42-2de71bdeab1f · Medium

The function $fx = 55.20 - 0.16x$ gives the estimated surface water temperature fx, in degrees Celsius, of a body of water on the xth day of the year, where $220 \leq x \leq 360$. Based on the model, what is the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year?

A. 55.20

B. 3.04

C. -0.16

D. -52.16

2.4 70799514-1e68-4ec1-9002-6b0e1165cbb0 · Medium

$$f(x) = 2x + 3$$

For the given function f, the graph of $y = f(x)$ in the xy-plane is parallel to line j. What is the slope of line j?

Student-produced response: _____

2.5 83a73190-62d2-4465-82e0-b6794ffbddf6 · Medium

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the y-intercept of the graph of $y = f(x)$ in the xy-plane in this context?

A. Caleb used approximately 5 fluid ounces of juice for each popsicle.

B. Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.

C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

D. Caleb used approximately 30 fluid ounces of juice for each popsicle.

2.6 5c630c81-9ef1-4180-821f-1dd4851610cf · Medium

In the linear function h, $h(28) = 15$ and $h(26) = 22$. Which equation defines h?

A. $h(x) = -\frac{2}{7}x + 23$

B. $h(x) = -\frac{2}{7}x + 113$

C. $h(x) = -\frac{7}{2}x + 23$

D. $h(x) = -\frac{7}{2}x + 113$

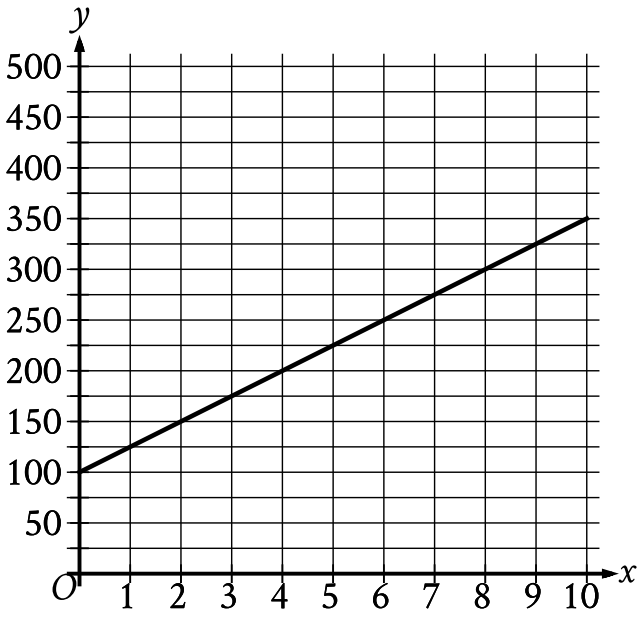
2.7 028813-DC · Medium

$c(x) = mx + 500$

A company's total cost $c(x)$, in dollars, to produce x shirts is given by the function above, where m is a constant and $x > 0$. The total cost to produce 100 shirts is $800. What is the total cost, in dollars, to produce 1000 shirts? (Disregard the $ sign when gridding your answer.)

Student-produced response: _____

2.8 d243de3a-1370-41b8-a350-5396b80263a0 · Medium



The graph of the function f, where $y = f(x)$, gives the total cost y, in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

A. Each game costs $\$25$.

B. The video game system costs $\$100$.

C. The video game system costs $\$25$.

D. Each game costs $\$100$.

2.9 47f92079-253b-46b3-a924-90dc197bbd5c · Medium

For the linear function h, the graph of $y = hx$ in the xy-plane passes through the points $7, 21$ and $9, 25$. Which equation defines h?

A. $hx = \frac{1}{2}x - \frac{7}{2}$

B. $hx = 2x + 7$

C. $hx = 7x + 21$

D. $hx = 9x + 25$

2.10 89816479-b500-4fe9-b80f-8f9fea007b9f · Medium

The table gives the number of hours, *h*, of labor and a plumber's total charge *fh*, in dollars, for two different jobs.

h	*fh*
1	155
3	285

There is a linear relationship between *h* and *fh*. Which equation represents this relationship?

A. $fh = 25h + 130$

B. $fh = 130h + 25$

C. $fh = 65h + 90$

D. $fh = 90h + 65$

2.11 f44d3920-a7ed-4c31-815c-11dbfe887d43 · Medium

In the *xy*-plane, the graph of the linear function *f* contains the points $0, 3$ and $7, 31$. Which equation defines *f*, where $y = fx$?

A. $fx = 28x + 34$

B. $fx = 3x + 38$

C. $fx = 4x + 3$

D. $fx = 7x + 3$

2.12 027788-DC · Medium

The function f is defined by $f(x) = mx + b$, where m and b are constants. If $f(0) = 18$ and $f(1) = 20$, what is the value of m ?

Student-produced response: _____

2.13 607e02b9-ac5b-48d2-8d14-a74f33743cd9 · Medium

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is $\$ 220$ when it sells 8 items, and its profit is $\$ 320$ when it sells 10 items. Which equation defines f?

A. $fx = 150x - 320$

B. $fx = 32x$

C. $fx = 50x - 10x$

D. $fx = 50x - 180$

2.14 718a67c3-595c-4af9-b1c7-38d2317665f0 · Medium

The owner of a bakery is giving away free muffins every hour. The equation $fx = 108 - 9x$ gives the number of free muffins, fx, that remain to be given away after x hours. Which of the following is the best interpretation of $f12 = 0$ in this context?

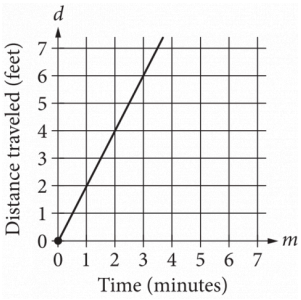
A. The owner of the bakery has 12 free muffins to give away.

B. The owner of the bakery will give away 12 free muffins in 9 hours.

C. It will take 12 hours to give away all the free muffins.

D. It will take 108 hours to give away 12 free muffins.

2.15 08375-DC · Medium



The graph above shows the distance traveled d, in feet, by a product on a conveyor belt m minutes after the product is placed on the belt. Which of the following equations correctly relates d and m ?

A. $d = 2m$

B. $d = \frac{1}{2}m$

C. $d = m + 2$

D. $d = 2m + 2$

2.16 6033a1e4-007d-42e9-ada3-25ad35a47e81 · Medium

In the *xy*-plane, line *k* has a slope of 5 and a *y*-intercept of $0, -35$. What is the *x*-coordinate of the *x*-intercept of line *k*?

Student-produced response: _____

2.17 32ff6707-691a-4b5c-a8ba-41a26d2c8ad0 · Medium

x	gx
1	32
2	28
3	24
4	20

For the linear function g, the table shows four values of x and their corresponding values of gx. The function can be written as $gx = mx + b$, where m and b are constants. What is the value of b?

A. 4

B. 16

C. 32

D. 36

2.18 ceae23d4-1965-4201-ac8f-3dbe916cc8e3 · Medium

In the linear function f, $f0 = 8$ and $f1 = 12$. Which equation defines f?

A. $fx = 12x + 8$

B. $fx = 4x$

C. $fx = 4x + 12$

D. $fx = 4x + 8$

2.19 10521-DC · Medium

Population of Greenleaf, Idaho

Year	Population
2000	862
2010	846

The table above shows the population of Greenleaf, Idaho, for the years 2000 and 2010. If the relationship between population and year is linear, which of the following functions P models the population of Greenleaf t years after 2000?

A. $P(t) = 862 - 1.6t$

B. $P(t) = 862 - 16t$

C. $P(t) = 862 + 16(t - 2{,}000)$

D. $P(t) = 862 - 1.6(t - 2{,}000)$

2.20 7fedd1ee-8dd3-449f-8f4a-8154d1d099ed · Medium

If $f(x) = x + 7$ and $g(x) = 7x$, what is the value of $4f(2) - g(2)$?

A. -5

B. 1

C. 22

D. 28

2.21 06a91641-79ae-4887-9807-53c2c9859f93 · Medium

$$w(t) = 300 - 4t$$

The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

A. 300

B. 296

C. 75

D. 4

2.22 a76b700a-7987-4650-b3bd-3b739ce06f18 · Medium

For the linear function h, $h0 = 0$ and $h11 = 10$. Which equation defines h?

A. $hx = \frac{10}{11}x$

B. $hx = \frac{10}{11}x + 10$

C. $hx = \frac{11}{10}x$

D. $hx = \frac{11}{10}x + 11$

2.23 d6701d8a-79c5-4cbd-b549-dcd4b570f33b · Medium

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

A. -4

B. 8

C. 42

D. 44

2.24 08691-DC · Medium

$y = 18 - 5x$

The equation above represents the speed y, in feet per second, of Sheila's bicycle x seconds after she applied the brakes at the end of a ride. If the equation is graphed in the xy-plane, which of the following is the best interpretation of the x-coordinate of the line's x-intercept in the context of the problem?

A. The speed of Sheila's bicycle, in feet per second, before Sheila applied the brakes

B. The number of feet per second the speed of Sheila's bicycle decreased each second after Sheila applied the brakes

C. The number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop

D. The number of feet Sheila's bicycle traveled from the time she began applying the brakes until the bicycle came to a complete stop

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2.25 022313-DC · Medium

$f(x) = \frac{(x+7)}{4}$

For the function f defined above, what is the value of $f(9) - f(1)$?

A. 1

B. 2

C. $\frac{1}{4}$

D. $\frac{9}{4}$

2.26 77e2c729-a66a-47de-9258-130399ad202e · Medium

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy-plane has an x-intercept at $(a, 0)$ and a y-intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

A. 21

B. 28

C. 32

D. 35

2.27 14854-DC · Medium

A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y, that remains to be moved x hours after the team started working.

$$y = 120 - 25x$$

The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?

A. The team will have moved all the cargo in about 4.8 hours.

B. The team has been moving about 4.8 tons of cargo per hour.

C. The team has been moving about 25 tons of cargo per hour.

D. The team started with 120 tons of cargo to move.

2.28 070898-DC · Medium

Energy per Gram of Typical Macronutrients

Macronutrient	Food calories	Kilojoules
Protein	4.0	16.7
Fat	9.0	37.7
Carbohydrate	4.0	16.7

The table above gives the typical amounts of energy per gram, expressed in both food calories and kilojoules, of the three macronutrients in food. If x food calories is equivalent to k kilojoules, of the following, which best represents the relationship between x and k ?

A. $k = 0.24x$

B. $k = 4.2x$

C. $x = 4.2k$

D. $xk = 4.2$

2.29 75be103e-3a39-4793-99d5-2ecb57774a84 · Medium

For the linear function g, the graph of $y = g(x)$ in the xy-plane has a slope of 2 and passes through the point $1, 14$. Which equation defines g?

A. $g(x) = 2x$

B. $g(x) = 2x + 2$

C. $g(x) = 2x + 12$

D. $g(x) = 2x + 14$

2.30 0478e0b5-4347-4ca3-81d3-9d8f3d89e005 · Medium

$$f(x) = 4x + b$$

For the linear function f, b is a constant and $f(7) = 28$. What is the value of b?

A. 0

B. 1

C. 4

D. 7

2.31 a60d5936-6770-4a18-b3b7-b21e68aadf5a · Medium

$$sx = 133 - 4x$$

The function s shown gives the number of spoons, sx, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s3 = 121$?

A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.

B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.

C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.

D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

2.32 022bc7aa-cc1a-41cc-8856-f626e8d243ac · Medium

Number of cars	Maximum number of passengers and crew
3	174
5	284
10	559

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A. $55c - p = -9$

B. $55c - p = 9$

C. $55p - c = -9$

D. $55p - c = 9$

2.33 940b3aad-7c94-4109-8fe4-ef6351e23009 · Medium

The relationship between two variables, x and y, is linear. For every increase in the value of x by 1, the value of y increases by 8. When the value of x is 2, the value of y is 18. Which equation represents this relationship?

A. $y = 2x + 18$

B. $y = 2x + 8$

C. $y = 8x + 2$

D. $y = 3x + 26$

2.34 0beb2470-2acd-4279-917c-0e0172947d97 · Medium

The function f is defined by $f(x) = -9x + 9$. What is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

Student-produced response: _____

2.35 3ad33584-7909-480c-b2c2-e9092550290d · Medium

In the xy-plane, the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$. Which equation defines f, where $y = f(x)$?

A. $f(x) = 2x + 42$

B. $f(x) = 32x + 36$

C. $f(x) = 4x + 2$

D. $f(x) = 8x + 2$

2.36 513fd8bd-86d1-4102-aea1-0a8bc65de5dd · Medium

$$f(x) = 39$$

For the given linear function f, which table gives three values of x and their corresponding values of fx?

A.

x	fx
0	0
1	0
2	0

B.

x	fx
0	39
1	39
2	39

C.

x	fx
0	0
1	39
2	78

D.

x	fx
0	39
1	0

2	-39

2.37 6bf54304-ff33-4fc3-8ab3-b9c8c115f618 · Medium

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers. Which equation could represent this linear relationship?

A. $t = 4d$

B. $t = \frac{1}{25}d$

C. $t = 25d$

D. $t = \frac{1}{4}d$

2.38 ee5be81f-8eae-4f80-8e45-e6642db5143d · Medium

x	gx
1	54
2	51
3	48
4	45

For the linear function g, the table shows four values of x and their corresponding values of gx. The function can be written as $gx = mx + b$, where m and b are constants. What is the value of b?

A. 3

B. 27

C. 54

D. 57

2.39 ef6e2e6b-9410-44c5-86c6-4cce2bdace80 · Medium

A model predicts that a certain animal weighed 241 pounds when it was born and that the animal gained 3 pounds per day in its first year of life. This model is defined by an equation in the form $fx = a + bx$, where fx is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. What is the value of a?

Student-produced response: _____

2.40 acd27116-7079-469e-b599-5c238ac3e563 · Medium

$$j(x) = mx + 144$$

For the linear function j, m is a constant and $j(12) = 18$. What is the value of $j(10)$?

Student-produced response: _____

2.41 ce40ca76-4f75-4034-9e39-b6c214f31eea · Medium

$$f(x) = 2x + 244$$

The given function f represents the perimeter, in centimeters (cm), of a rectangle with a length of x cm and a fixed width. What is the width, in cm, of the rectangle?

A. 2

B. 122

C. 244

D. 488

2.42 025468-DC · Medium

In the xy-plane, the points $(-2,3)$ and $(4,-5)$ lie on the graph of which of the following linear functions?

A. $f(x) = x + 5$

B. $f(x) = \frac{1}{2}x + 4$

C. $f(x) = -\frac{4}{3}x + \frac{1}{3}$

D. $f(x) = -\frac{3}{2}x + 1$

2.43 9a75850f-42ca-4792-94aa-fe7d5258a84c · Medium

$$f(x) = 45x + 600$$

The function f gives the monthly fee $f(x)$, in dollars, a facility charges to keep x crates in storage. What is the monthly fee, in dollars, the facility charges to keep 50 crates in storage?

Student-produced response: _____

2.44 05013-DC · Medium

x	$f(x)$
0	−2
2	4
6	16

Some values of the linear function f are shown in the table above. What is the value of $f(3)$?

A. 6

B. 7

C. 8

D. 9

2.45 9a64af20-3d37-416c-9771-0ce29547dd42 · Medium

In the *xy*-plane, line *k* passes through the points $8, 6$ and $9, 12$ and is defined by the equation $y = mx + b$. What is the value of *b*?

Student-produced response: _____

2.46 029018-DC · Medium

The boiling point of water at sea level is 212 degrees Fahrenheit $(°F)$. For every 550 feet above sea level, the boiling point of water is lowered by about $1°F$. Which of the following equations can be used to find the boiling point B of water, in $°F$, x feet above sea level?

A. $B = 550 + \frac{x}{212}$

B. $B = 550 - \frac{x}{212}$

C. $B = 212 + \frac{x}{550}$

D. $B = 212 - \frac{x}{550}$

2.47 0d590b1e-c165-452b-bd68-2c7ae4a0a55d · Medium

The function f is defined by $f(x) = 4x + kx - 1$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

Student-produced response: _____

2.48 070611-DC · Medium

According to a model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to four times the body weight of the bee, in grams. According to the model, what would be the head width, in millimeters, of a worker bumblebee that has a body weight of 0.5 grams?

Student-produced response: _____

2.49 f0bbaf49-87a8-4fb3-b54f-eadea2dca1a9 · Medium

The function f is defined by $f(x) = 7x - 84$. What is the x-intercept of the graph of $y = f(x)$ in the xy-plane?

A. $(-12, 0)$

B. $(-7, 0)$

C. $(7, 0)$

D. $(12, 0)$

2.50 74ab7488-2efa-4e42-86ec-970c72177393 · Medium

Brian saves $\frac{2}{5}$ of the $\$ 215$ he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in 9 weeks?

Student-produced response: _____

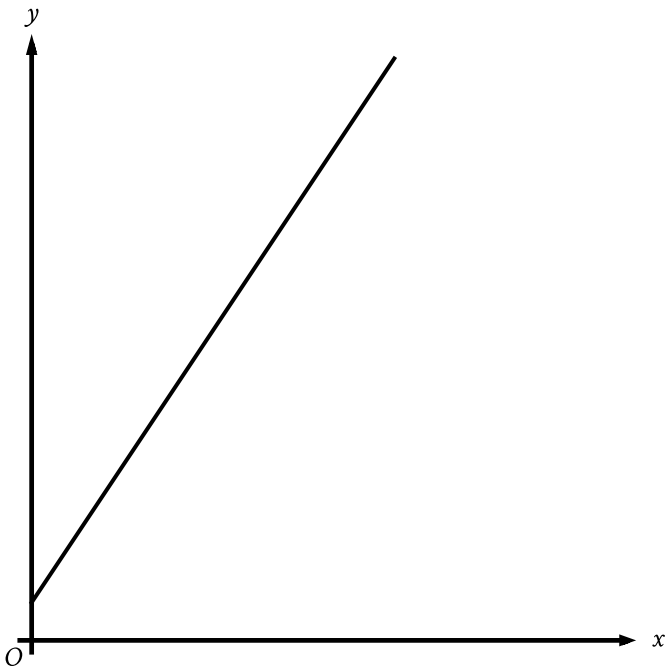
2.51 b64bd9d3-56b1-45f6-b301-f29dfd4eb579 · Medium

x	10	15	20	25
fx	82	137	192	247

The table shows four values of x and their corresponding values of fx. There is a linear relationship between x and fx that is defined by the equation $fx = mx - 28$, where m is a constant. What is the value of m?

Student-produced response: _____

2.52 461f9ba1-a36e-4795-9b6e-093390fd8977 · Medium



- The line begins on the y axis above the x axis.
- The line slants sharply up from left to right.

The graph represents the total charge, in dollars, by an electrician for *x* hours of work. The electrician charges a onetime fee plus an hourly rate. What is the best interpretation of the slope of the graph?

A. The electrician’s hourly rate

B. The electrician’s onetime fee

C. The maximum amount that the electrician charges

D. The total amount that the electrician charges

2.53 11b06370-1f90-4553-8b54-c1ebec99dcad · Medium

$$g(x) = 314x - 15$$

What is the y-coordinate of the y-intercept of the graph of $y = g(x) - 2$ in the xy-plane?

A. -47

B. -45

C. -17

D. -15

2.54 f268654f-d968-4474-abc8-911195b0c902 · Hard

For the function f, $f(c)(x) = x - 8$ for all values of x, where c is a positive constant. If $f(2) = 35$, what is the value of c?

Student-produced response: _____

2.55 e47cdf24-1ef9-4aa4-bc46-70b25da6e45e · Hard

The equation $h = \frac{9v - 273.15}{5} + 32$ gives the corresponding temperature h, in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of 467.33 degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

Student-produced response: _____

2.56 5bba5291-1a05-43fc-bef6-75ab59bb1c60 · Hard

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is $\$ 25.40$, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

A. $fx = 25.40(x - 1) + 12.70$

B. $fx = 25.40(x - 2) + 12.70$

C. $fx = 25.40x - 1 + 12.70x - 2$

D. $fx = 25.40x - 2 + 12.70x - 1$

2.57 025812-DC · Hard

An economist modeled the demand Q for a certain product as a linear function of the selling price P. The demand was 20,000 units when the selling price was $40 per unit, and the demand was 15,000 units when the selling price was $60 per unit. Based on the model, what is the demand, in units, when the selling price is $55 per unit?

A. 16,250

B. 16,500

C. 16,750

D. 17,500

2.58 1a8089dd-7f49-40bb-9cf4-9d9040daa84f · Hard

A window repair specialist charges $\$ 220$ for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is $\$ 400$. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

2.59 021098-DC · Hard

For the function f, if $f(3x) = x - 6$ for all values of x, what is the value of $f(6)$?

A. -6

B. -4

C. 0

D. 2

2.60 f3a8f6e5-3640-4af9-a194-e13afbea670e · Hard

x	*fx*
1	-64
2	0
3	64

For the linear function *f*, the table shows three values of *x* and their corresponding values of *fx*. Function *f* is defined by *fx* = *ax* + *b*, where *a* and *b* are constants. What is the value of *a* - *b*?

A. -64

B. 62

C. 128

D. 192

2.61 4c14e865-e027-430a-9613-c80acd0346bf · Hard

For a particular car, the linear function f gives the predicted power, in brake horsepower bhp, for engine speeds between $1{,}000$ revolutions per minute rpm and $6{,}000$ rpm. According to this function, the car's predicted power is 433 bhp at an engine speed of $3{,}331$ rpm and 600 bhp at an engine speed of $4{,}500$ rpm. The equation $fx = \frac{1}{7}x - a + 433$ defines f, where x is the engine speed, in rpm, and a is a constant. What is the value of a?

Student-produced response: _____

2.62 6ea75b24-df55-4b56-a799-54ee041ec141 · Hard

The cost of renting a carpet cleaner is $\$ 52$ for the first day and $\$ 26$ for each additional day. Which of the following functions gives the cost Cd, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

A. $C(d) = 26d + 26$

B. $C(d) = 26d + 52$

C. $C(d) = 52d - 26$

D. $C(d) = 52d + 78$

2.63 9757792c-41ae-4f55-afe7-6d7d77147ac6 · Hard

A train traveled 173 miles in the first 3 hours of a trip and traveled at an average speed of 53 miles per hour for the remainder of the trip. Which equation gives the total number of miles, y, the train traveled if the trip was x hours long and $x > 3$?

A. $y = 53x + 173$

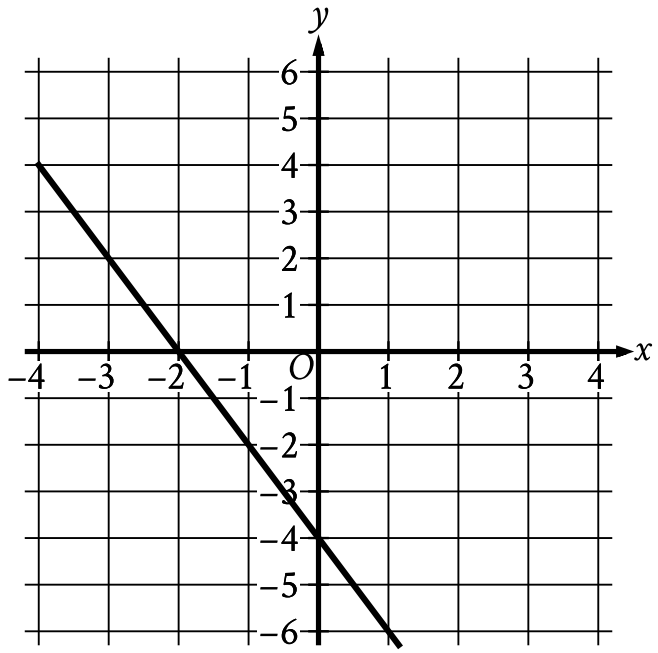
B. $y = 53x + 14$

C. $y = 14x + 226$

D. $y = 14x + 53$

2.64 c0070404-51d0-404d-88ce-75aa7e687a20 · Hard

The graph of $y = fx - 11$ is shown.



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (negative 1 comma negative 2)
  - (0 comma negative 4)

Which equation defines the linear function f?

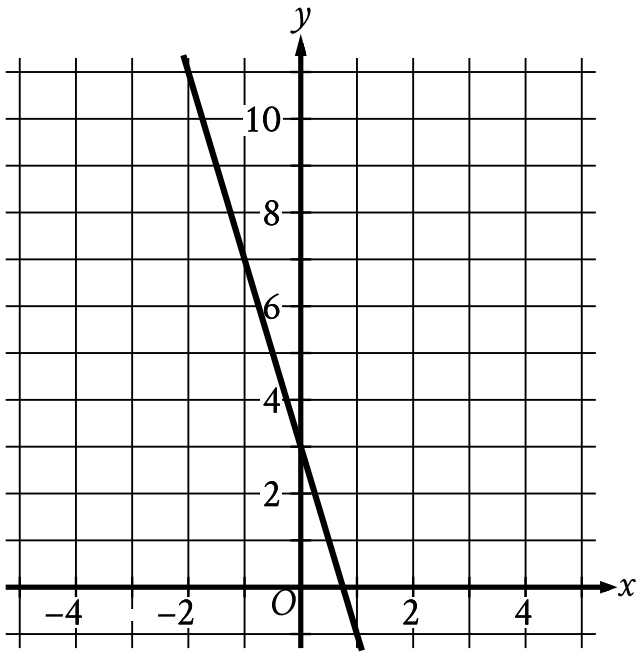
A. $fx = -13x - 11$

B. $fx = -2x + 7$

C. $fx = -13x + 7$

D. $fx = -2x - 11$

2.65 2d19a50f-046d-44a2-92f8-a10cf81a3e3c · Hard



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (negative 1 comma 7)
  - (0 comma 3)

The graph of the linear function $y = fx + 19$ is shown. If c and d are positive constants, which equation could define f?

A. $fx = -d - cx$

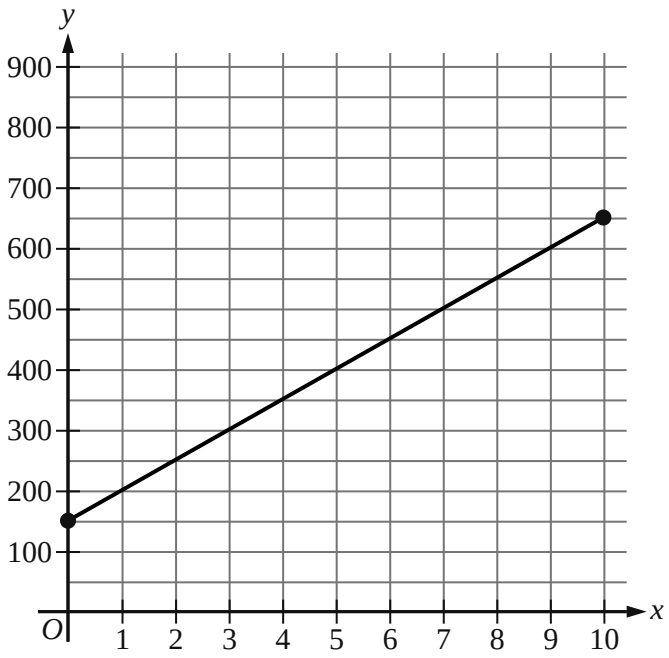
B. $fx = d - cx$

C. $fx = -d + cx$

D. $fx = d + cx$

2.66 dfe2986a-18ec-40e9-8bd5-2f33f9975a35 · Hard

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance *y*, in meters, Mary had walked on the trail *x* minutes after leaving the trail sign.



- The line segment slants gradually up from left to right.
- The line segment begins at the point (0 comma 150).
- The line segment ends at the point (10 comma 650).

What distance, in meters, did Mary walk on the trail in the 10 minutes after leaving the trail sign?

Student-produced response: _____

2.67 1f9e0de2-d5a7-4f67-9e59-62315da247f3 · Hard

A linear model estimates the population of a city from 1991 to 2015. The model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and *x* thousand in 2015. To the nearest whole number, what is the value of *x*?

Student-produced response: _____

2.68 bb4ec6e5-ecea-4bc4-a6d2-40a50bb5e60f · Hard

$$Fx = \frac{9}{5}x - 273.15 + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 2.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A. 3.78

B. 35.78

C. 487.89

D. 519.89

2.69 09384-DC · Hard

Oil and gas production in a certain area dropped from 4 million barrels in 2000 to 1.9 million barrels in 2013. Assuming that the oil and gas production decreased at a constant rate, which of the following linear functions f best models the production, in millions of barrels, t years after the year 2000?

A. $f(t) = \frac{21}{130}t + 4$

B. $f(t) = \frac{19}{130}t + 4$

C. $f(t) = -\frac{21}{130}t + 4$

D. $f(t) = -\frac{19}{130}t + 4$

2.70 029517-DC · Hard

x	−11	−10	−9	−8
f(x)	21	18	15	12

The table above shows some values of x and their corresponding values $f(x)$ for the linear function f. What is the x-intercept of the graph of $y = f(x)$ in the xy-plane?

A. $(-3,0)$

B. $(-4,0)$

C. $(-9,0)$

D. $(-12,0)$

2.71 ff095f0f-b8e2-470c-903c-78999ec17213 · Hard

The cost of renting a large canopy tent for up to 10 days is $ 430 for the first day and $ 215 for each additional day. Which of the following equations gives the cost *y*, in dollars, of renting the tent for *x* days, where *x* is a positive integer and $x \leq 10$?

A. $y = 215x + 215$

B. $y = 430x - 215$

C. $y = 430x + 215$

D. $y = 215x + 430$

2.72 700cd854-082d-4c3c-a2da-017482d48062 · Hard

One gallon of paint will cover 220 square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P, in gallons, needed to paint the walls of the room twice?

A. $P = \frac{w}{110}$

B. $P = 440w$

C. $P = \frac{w}{220}$

D. $P = 220w$

2.73 704bbab2-6312-41d6-a1ca-affd16b16c5a · Hard

The linear function g is defined by $gx = b - 15x$, where b is a constant. If $gc + 7 = \frac{c}{4}$, where c is a constant, which of the following expressions represents the value of b?

A. $\frac{15c}{4}$

B. $\frac{19c}{4} + 7$

C. $\frac{61c}{4} + 105$

D. $15c + 105$

2.74 09335-DC · Hard

According to data provided by the US Department of Energy, the average price per gallon of regular gasoline in the United States from September 1, 2014, to December 1, 2014, is modeled by the function F defined below, where $F(x)$ is the average price per gallon x months after September 1.

$$F(x) = 2.74 - 0.19(x - 3)$$

The constant 2.74 in this function estimates which of the following?

A. The average monthly decrease in the price per gallon

B. The difference in the average price per gallon from September 1, 2014, to December 1, 2014

C. The average price per gallon on September 1, 2014

D. The average price per gallon on December 1, 2014

2.75 ea18109c-11ea-4e1e-a907-b4a77b4a93e1 · Hard

$$Fx = \frac{9}{5}x - 273.15 + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 9.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A. 16.38

B. 48.38

C. 475.29

D. 507.29

2.76 839b7fb0-37a1-46d9-9dce-f3af7e85c97e · Hard

The functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x-coordinate of the x-intercept of the graph of $y = hx$ in the xy-plane?

Student-produced response: _____

2.77 4959c26d-16dc-4df2-8bb7-92bd72d87aa5 · Hard

For groups of 25 or more people, a museum charges $\$ 21$ per person for the first 25 people and $\$ 14$ for each additional person. Which function f gives the total charge, in dollars, for a tour group with n people, where $n \geq 25$?

A. $fn = 14n + 175$

B. $fn = 14n + 525$

C. $fn = 35n - 350$

D. $fn = 14n + 21$

2.78 d70b4faa-5ce1-4664-ae31-432dc125273a · Hard

The cost of renting a backhoe for up to 10 days is $\$ 270$ for the first day and $\$ 135$ for each additional day. Which of the following equations gives the cost y, in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

A. $y = 270x - 135$

B. $y = 270x + 135$

C. $y = 135x + 270$

D. $y = 135x + 135$

2.79 5904d2e1-1ae5-40cf-915f-050b6b6d3111 · Hard

One gallon of stain will cover 170 square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S, in gallons, needed to stain the fence in this yard twice?

A. $S = \frac{w}{170}$

B. $S = 170w$

C. $S = 340w$

D. $S = \frac{w}{85}$

2.80 b56fbc38-168f-4197-8169-fd6b32162d42 · Hard

x	y
-34	t
-17	$t + 23$
0	$t + 46$

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y, where t is a constant. Which equation represents this relationship?

A. $y = -2x + t + 23$

B. $y = 2x + t + 23$

C. $y = -\frac{23}{17}x + t + 46$

D. $y = \frac{23}{17}x + t + 46$

2.81 08486-DC · Hard

An object hangs from a spring. The formula $\ell = 30 + 2w$ relates the length ℓ, in centimeters, of the spring to the weight w, in newtons, of the object. Which of the following describes the meaning of the 2 in this context?

A. The length, in centimeters, of the spring with no weight attached

B. The weight, in newtons, of an object that will stretch the spring 30 centimeters

C. The increase in the weight, in newtons, of the object for each one-centimeter increase in the length of the spring

D. The increase in the length, in centimeters, of the spring for each one-newton increase in the weight of the object

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2.82 b09c918e-97c2-4a8b-9406-9ed217790fb4 · Hard

x	*y*
-12	-45
6	45

The table shows two values of *x* and their corresponding values of *y*. The graph of the linear equation representing this relationship passes through the point $\frac{1}{4}$, a. What is the value of *a*?

Student-produced response: _____

2.83 4e317fbb-cdce-4c3a-acc2-5f1f342d3af0 · Hard

Kaylani used fabric measuring 5 yards in length to make each suit for a men's choir. The relationship between the number of suits that Kaylani made, x, and the total length of fabric that she purchased y, in yards, is represented by the equation $y - 5x = 6$. What is the best interpretation of 6 in this context?

A. Kaylani made 6 suits.

B. Kaylani purchased a total of 6 yards of fabric.

C. Kaylani used a total of 6 yards of fabric to make the suits.

D. Kaylani purchased 6 yards more fabric than she used to make the suits.

2.84 b054f030-a6ac-4c84-8808-fddfd6d758ba · Hard

The function $f(x)$ is defined as 19 more than 4 times a number x. If $y = f(x)$ is graphed in the xy-plane, what is the best interpretation of the x-intercept?

A. When $f(x) = 0$, the number is $-\frac{19}{4}$.

B. When the number is 0, $f(x) = 19$.

C. The value of $f(x)$ increases by 1 for each increase of 4 in the value of the number.

D. For each increase of 1 in the value of the number, $f(x)$ increases by 4.

2.85 c16f36d3-645b-4e76-856d-2cc2c923b53c · Hard

x	fx
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For the linear function f, the table shows three values of x and their corresponding values of fx. If $hx = fx - 13$, which equation defines h?

A. $hx = 5x - 4$

B. $hx = 5x + 7$

C. $hx = 5x + 9$

D. $hx = 5x + 20$